Amount to be paid = 3.75 × 62.50

= ₹234.3750

= ₹234.38 (rounded to two decimals).

	62.50	← 2 decimal place
×	3.75	← 2 decimal place
	31250	
	4375	
	1875	
	234.3750	← 4 decimal place(s)

Example 1.20 The length and breadth of a rectangle is 23.5 *cm* and 1.5 *cm* respectively. Find the area of the rectangle.

Solution

Area of a rectangle = $l \times b$ *sq. units*

Here, l=23.5 *cm*, b=1.5 *cm*

Area of the rectangle = 23.5 × 1.5

= 35.25 *sq.cm.*

	23.5	← 1 decimal place
×	1.5	← 1 decimal place
	1175	
	235	
	35.25	← 2 decimal place(s)

(ii) Multiplication of Decimal Numbers by 10, 100 and 1000

We have studied about conversion of decimals into fractions in the first term. Consider 45.6 and 4.56.

Expressing these decimal numbers into fractions, we get

$$45.6=40+5+\frac{6}{10}=45+\frac{6}{10}=\frac{456}{10}$$

Now, $4.56=4+\frac{5}{10}+\frac{6}{100}=\frac{456}{100}$

Comparing the two fractions, we see that if there is one digit after the decimal point, then the denominator is 10 and if there are two digits after the decimal point, then the denominator is 100.

Let us see what happens if a decimal number is multiplied by 10, 100 and 1000.

Observe the following table and complete it.

Try these

$2.35\times10=\frac{235}{100}\times10=\frac{235}{10}=23.5$	7.63×10 = _____	63.237×10 =_____
$2.35\times100=\frac{235}{100}\times100=235=235.0$	7.63×100 = _____	63.237×100 =_____
$2.35\times1000=\frac{235}{100}\times1000=2350.0$	7.63×1000 =_____	63.237×1000 =_____
$0.6\times10=\frac{6}{10}\times10=6$	$0.6\times100=\frac{6}{10}\times100$ =_____	$0.6\times1000=\frac{6}{10}\times1000$ =_____

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